

QUESTION:

Objective

To understand and implement the basic programming constructs of Java, including variables, data types, operators, conditional statements, loops, and methods.

Concepts Covered

- Java Program Structure
- Variables and Data Types
- Operators
- Input and Output
- Conditional Statements (if, if-else, switch)
- Looping Statements (for, while, do-while)
- Methods (User-defined Functions)

Task Description

Develop a "**Student Utility Program**" that performs the following operations using a menu-driven approach. Part A: Java Basics

- Display student information.
- Accept Name, Roll Number, and Marks.
- Calculate Total and Percentage.
- Display the result.

Part B: Conditional Statements

Implement menu options to:

- Check whether a number is Even or Odd.
- Find the Largest of Three Numbers.
- Display Grade based on percentage using if-else.
- Display Day of the Week using switch.

Part C: Looping Statements

Write programs to:

- Print multiplication tables.
- Display numbers from 1 to N.
- Calculate the sum of first N natural numbers.
- Generate the Fibonacci series.

Part D: Methods : Every method should be parameterized and should return the result:

Create user-defined methods for:

- Finding the factorial of a number.
- Checking whether a number is prime.
- Finding the maximum of two numbers.
- Calculating the area of a circle.

Call each method from the main() method.

Step wise execution

Step 1: Start the Java program.

Step 2: Import the Scanner class to accept input from the user.

Step 3: Display the Student Utility Program menu.

Step 4: Ask the user to enter a menu choice.

Step 5: If the choice is 1, accept student name, roll number and marks, calculate total and percentage, and display the result.

Step 6: If the choice is 2, accept a number and check whether it is even or odd using the if-else statement.

Step 7: If the choice is 3, accept three numbers and find the largest using if-else.

Step 8: If the choice is 4, accept percentage and display the grade using if-else.

Step 9: If the choice is 5, accept a number from 1 to 7 and display the corresponding day using switch.

Step 10: If the choice is 6, accept a number and display its multiplication table from 1 to 10.

Step 11: If the choice is 7, accept N and display numbers from 1 to N.

Step 12: If the choice is 8, accept N and calculate the sum of the first N natural numbers.

Step 13: If the choice is 9, accept the number of terms and generate the Fibonacci series.

Step 14: If the choice is 10, call the factorial() method and display the returned result.

Step 15: If the choice is 11, call the prime() method and check whether the number is prime.

Step 16: If the choice is 12, call the maximum() method and display the maximum of two numbers.

Step 17: If the choice is 13, call the areaCircle() method and display the area of the circle.

Step 18: If the choice is 14, terminate the program.

Step 19: Repeat the menu until the user selects Exit.

Step 20: Stop the program.

Algorithm

1. Start.
2. Create a Scanner object for user input.
3. Display the menu.
4. Read the user's choice.
5. Perform the selected operation:

a) Student Information:

- i. Read name, roll number and three marks.
- ii. Calculate total = mark1 + mark2 + mark3.
- iii. Calculate percentage = total / 3.
- iv. Display student details.

b) Even or Odd:

- i. Read a number.
- ii. If number % 2 == 0, display Even.
- iii. Otherwise display Odd.

c) Largest of Three:

- i. Read three numbers.
- ii. Compare the numbers using if-else.
- iii. Display the largest number.

d) Grade:

- i. Read percentage.
- ii. Use if-else conditions.
- iii. Display the appropriate grade.

e) Day of Week:

- i. Read day number.
- ii. Use switch.
- iii. Display the corresponding day.

f) Multiplication Table:

- i. Read a number.
- ii. Use a for loop from 1 to 10.
- iii. Display the multiplication table.

g) Numbers 1 to N:

- i. Read N.
- ii. Use a for loop.
- iii. Display numbers from 1 to N.

h) Sum of N Natural Numbers:

- i. Read N.
- ii. Initialize sum = 0.
- iii. Add numbers from 1 to N.
- iv. Display sum.

i) Fibonacci Series:

- i. Read number of terms.
- ii. Initialize first = 0 and second = 1.
- iii. Generate the next terms using a loop.
- iv. Display the series.

j) Factorial:

- i. Read a number.
- ii. Call factorial().
- iii. Display the returned result.

k) Prime:

- i. Read a number.
- ii. Call prime().
- iii. Display whether the number is prime.

l) Maximum:

- i. Read two numbers.
- ii. Call maximum().
- iii. Display the returned maximum.

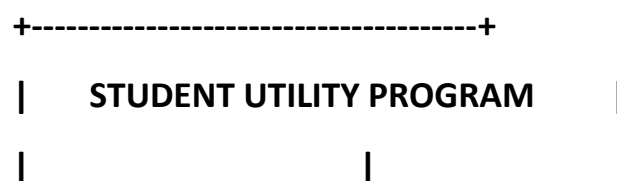
m) Area of Circle:

- i. Read radius.
- ii. Call areaCircle().
- iii. Display the returned area.

6. Repeat steps 3–5 until choice = 14.

7. Stop.

UML diagram



(Student Information)	
(Even or Odd)	
(Largest of Three)	
(Calculate Grade)	
(Display Day)	
(Multiplication Table)	
(Display 1 to N)	
(Sum of Natural Numbers)	
(Fibonacci Series)	
(Find Factorial)	
(Check Prime)	
(Find Maximum)	
(Calculate Circle Area)	
(Exit)	

+-----+	
+-----+	
User	
+-----+	

Code:

```
package basics;

import java.util.Scanner;
```

```
public class StudentUtility {
    // Factorial method
    static int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }
    // Prime Number method
    static boolean Prime(int n) {
        if (n <= 1)
            return false;
        for (int i = 2; i < n; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }
    // Maximum method
    static int maximum(int a, int b) {
        if (a > b)
            return a;
        else
            return b;
    }
    // Area of circle method
    static int area(int r) {
        return (int)(3.14 * r * r);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        // Part A: Student Details
        System.out.println("Student details");
        System.out.println("Enter a student name: ");
    }
}
```



```

String name = sc.next();
System.out.println("Enter student roll no: ");
int rollno = sc.nextInt();
System.out.println("Enter student marks: ");
int[] marks = new int[5];
int total = 0;
for (int i = 0; i < 5; i++) {
    marks[i] = sc.nextInt();
    total = total + marks[i];
}
double percent = (double) total / marks.length;
System.out.println("Percentage: " + percent);
System.out.println("Total marks: " + total);
// Grade
if (percent >= 90)
    System.out.println("Grade A++");
else if (percent >= 80)
    System.out.println("Grade A");
else if (percent >= 70)
    System.out.println("Grade B");
else if (percent >= 60)
    System.out.println("Grade C");
else
    System.out.println("Fail in Exam");
int choice;
do {
    System.out.println("\nMenu");
    System.out.println(" 1. Even or odd\n2. Largest of
three numbers\n3. Day of week\n4. Multiplication table\n5.
Numbers (1 to N)");
    System.out.println("6. Sum of N natural numbers\n7.
Fibonacci series\n8. Method\n9. Exit");

    System.out.println("Enter choice: ");
    choice = sc.nextInt();

```

```
switch (choice) {

    case 1:
        System.out.println("Enter n: ");
        int n = sc.nextInt();

        if (n % 2 == 0)
            System.out.println("Even number");
        else
            System.out.println("Odd number");

        break;

    case 2:
        System.out.println("Enter 3 numbers: ");

        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a > b && a > c)
            System.out.println(a + " is larger
number");
        else if (b > a && b > c)
            System.out.println(b + " is larger
number");
        else
            System.out.println(c + " is larger
number");

        break;

    case 3:
        System.out.println("Enter day: ");
        int day = sc.nextInt();
```

```

        switch (day) {
            case 1:
                System.out.println("Sunday");
                break;
            case 2:
                System.out.println("Monday");
                break;
            case 3:
                System.out.println("Tuesday");
                break;
            case 4:
                System.out.println("Wednesday");
                break;
            case 5:
                System.out.println("Thursday");
                break;
            case 6:
                System.out.println("Friday");
                break;
            case 7:
                System.out.println("Saturday");
                break;
            default:
                System.out.println("Invalid day");
        }
        break;
    case 4:
        System.out.println("Enter multiplication
table: ");

        int x = sc.nextInt();
        for (int i = 0; i <= 20; i++) {
            System.out.println(x + " * " + i + " =
" + (x * i));
        }
        break;
    case 5:

```

```

        System.out.println("Enter Natural number
(N): ");

        int y = sc.nextInt();
        for (int i = 0; i <= y; i++) {
            System.out.println(i);
        }
        break;
    case 6:
        System.out.println("Enter N number: ");
        int z = sc.nextInt();
        int sum = 0;
        for (int i = 0; i <= z; i++) {
            sum += i;
        }
        System.out.println("Sum of N natural
numbers: " + sum);
        break;
    case 7:
        System.out.println("Enter Fibonacci series
(N): ");

        int N= sc.nextInt();
        int a1 = 0;
        int b1 = 1;
        for (int i = 0; i < N; i++) {
            System.out.print(a1 + " ");
            int next = a1 + b1;
            a1 = b1;
            b1 = next;
        }
        break;
    case 8:
        System.out.println("1. Factorial\n2.
Prime\n3. Maximum\n4. Area\nEnter method choice: ");
        int method = sc.nextInt();
        switch (method) {
            case 1:

```

```

        int f = factorial(10);
        System.out.println("Factorial is "
+ f);

        break;
    case 2:
        boolean p = Prime(5);
        System.out.println("Prime is " +
p);

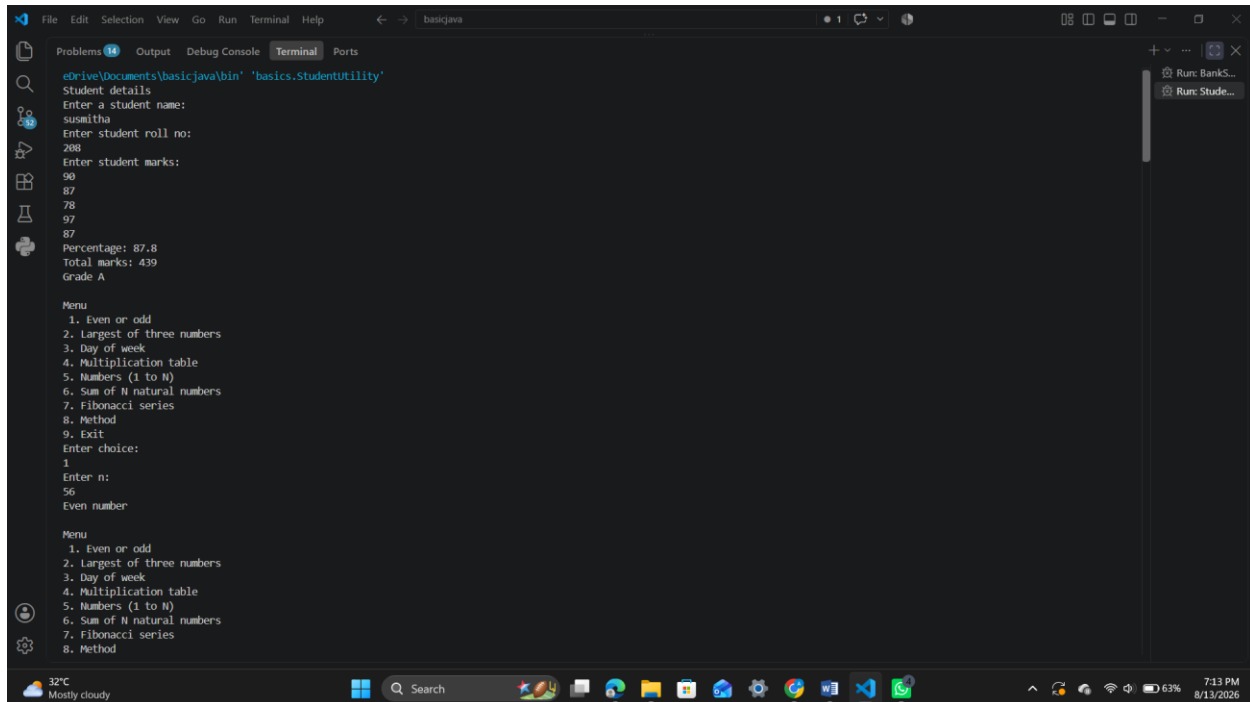
        break;
    case 3:
        int M = maximum(14, 19);
        System.out.println("Maximum number
is " + M);

        break;
    case 4:
        double result = area(3);
        System.out.println("Area of circle:
" + result);

        break;
    default:
        System.out.println("No method");
    }
    break;
case 9:
    System.exit(0);
    break;
default:
    System.out.println("Invalid number");
}
} while (choice != 0);
sc.close();
}
}

```

Output screen shot



```
File Edit Selection View Go Run Terminal Help
eDrive\Documents\basicjava\bin\ 'basics.StudentUtility'
Student details
Enter a student name:
susmitha
Enter student roll no:
208
Enter student marks:
90
87
78
97
87
Percentage: 87.8
Total marks: 439
Grade A

Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
1
Enter n:
56
Even number

Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
```

```
Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
3
Enter day:
5
Thursday

Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
4
Enter multiplication table:
34
34 * 0 = 0
34 * 1 = 34
34 * 2 = 68
34 * 3 = 102
34 * 4 = 136
34 * 5 = 170
34 * 6 = 204
34 * 7 = 238
34 * 8 = 272
34 * 9 = 306
34 * 10 = 340
```

```
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
6
Enter N number:
54
Sum of N natural numbers: 1485

Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
7
Enter Fibonacci series (N):
9
0 1 1 2 3 5 8 13 21

Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
8
1. Factorial
2. Prime
3. Maximum
4. Area
```

```
File Edit Selection View Go Run Terminal Help
basicjava
Problems 14 Output Debug Console Terminal Ports
Enter Natural number (N):
4
0
1
2
3
4
Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
8
1. Factorial
2. Prime
3. Maximum
4. Area
Enter method choice:
4
Area of circle: 28.0

Menu
1. Even or odd
2. Largest of three numbers
3. Day of week
4. Multiplication table
5. Numbers (1 to N)
6. Sum of N natural numbers
7. Fibonacci series
8. Method
9. Exit
Enter choice:
9
PS C:\Users\Admin\OneDrive\Documents\basicjava>
```